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REGULAR ARTICLE

Gender Gap Among High Achievers in Math and Implications for STEM Pipeline

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Abstract This study examined a new form of pervasive gender inequality: the gender gap among high achievers in math and considered its implication for developing STEM talents. Using the cross-nation Programme for International Student Achievement (PISA) data from both 2003 and 2012, we examined the mathematics gender gap among 15-year-old high achievers across ten countries/regions. We showed a consistent male advantage among the top performers in mathematics. Follow-up regression analyses revealed that the gap was associated with some socio-demographic and schooling/attitudinal variables, even after controlling for the background variables. We argue that education communities should acknowledge and address this form of gender inequality, as it could have ramifications for the science, technology, engineering, and mathematics (STEM) education pipeline. Educators and society in general still face challenges in closing the gender gap among high achievers in math with an aim to develop a gender-balanced STEM talent pool.

Keywords Gender differences · Mathematics · Achievement · STEM · PISA

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Introduction

Gender inequality in educational outcomes has been a hotly contested issue in academic debates. While male students have historically enjoyed a large performance edge in mathematics, more recently, mounting evidence has documented a trend that such a gap has been on the decline (Lindberg et al. 2010). More recent findings from large-scale assessment data typically reveal a slight male advantage in average math achievement (Langenfeld 1997; Li 1999; Liu and Wilson 2009; Mullis et al. 2000; Mullis et al. 2005; National Center for Education Statistics 2002, 2005; OECD 2004). Additional recent research findings further suggest that the math gender gap in the form of gender group mean difference has decreased to the level that it is no longer consequential, thus leading to the hypothesis of "gender similarities in math achievement" (Hyde 2005; Hyde et al. 2008; Hyde and Mertz 2009).

Alarmingly, although the average gender gap in math performance may have shrunk, this trend has yet to alleviate the gender gap still prevalent in the STEM fields (Science, Technology, Engineering, and Mathematics). Data from the National Science Foundation (2008) in the United States showed that, relative to their male counterparts, considerably fewer females attained degrees in STEM fields where mathematical preparation is essential. Similar phenomenon can also be found in other industrialized or developing countries (World Economic Forum 2015).

In addition to the aforementioned conflicting trends in gender achievement within STEM education, researchers began to notice a different manifestation of the gender gap: student distribution by gender at particular performance benchmarks (Guiso et al. 2008; Hyde and Mertz 2009; Penner 2008; Pennar and Paret 2008; Stoet and Geary

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2013). It is possible that the mean achievement only tells a part of the story. Existing evidence suggests that the converging average performance could be accompanied by a large dispersion on the performance spectrum. There could be considerably fewer female high achievers in math than males. Educators have long been aware that successful STEM students typically have a strong command of mathematics, in other words, they are more likely to be represented in the upper tail of the distribution. Because of this issue, our study aims to develop a better understanding of the gender difference at the upper end of the math performance distribution as another form of gender inequality. Using international assessment data, we look at the math performance for ten education systems and examined some factors that could contribute to the differential gender distribution patterns among high achievers in mathematics. We discuss this issue in the context of STEM education.

Literature Review

Gender Gap in Mathematics

Studies on gender difference in math typically rely on the gender mean difference. Only a few studies have examined the gender gap near the tails of math performance distributions (Guiso et al. 2008; Hyde and Mertz 2009; Penner 2008; Pennar and Paret 2008; Stoet and Geary 2013). As a general societal concern, education equity between gender groups is undeniably important in understanding the average performance difference between gender groups. However, this perspective of gender group mean difference has its limits, as it is not able to depict a full picture of gender-related inequality in mathematics.

Current research on gender difference at the extreme tails of performance distribution is emerging, although such research only offers an incomplete picture at this time. Some researchers examined the hypothesis of greater male variability (Machin and Pekkarinen 2008), and the gender difference near the upper tail of math performance distribution (Hyde 2005; Hyde et al. 2008; Hyde and Mertz 2009). Hyde et al. (2008), among other things, considered the issue of gender imbalance among the high achievers in math using US state assessment data. They found that among white students, the male-to-female ratio above the 95th performance percentile was 1.45, and the ratio above the 99th percentile was 2.06 (Hyde et al. 2008, p. 494). However, for Asian-Americans, the ratio above the 95th percentile was 1.09, but reversed to 0.91 for those who were above the 99th percentile. Stoet and Geary (2013) used PISA data to estimate the gender ratio at the extreme end. By combining all countries together, they showed the male-to-female ratio ranged from 1.6 to 1.9 above the 95th percentile, and from 2.1 to 2.6 above the 99th percentile, across several waves of PISA data. Machin and Pekkarinen (2008) analyzed PISA 2003 data country by country, and revealed that 39 out of 41 countries showed a larger-than-one male-to-female ratio above the 95th percentile of math performance distribution, indicating a male advantage among the high-performing students in math. Though countries have increasingly participated in the international assessments, allowing for more waves of data to be accumulated, there have not been any recent attempts to study the gender gap at the high tails of the math performance distribution.

Implications for the STEM Pipeline

STEM education has become a focus for many global education reforms. For instance, in the United States, the National Science Foundation teamed up with higher education institutions to implement innovative STEM programs under its flagship grant, Improving Undergraduate STEM Education (IUSE). For younger children at the secondary level or lower, the STEM Education Initiative provides various resources and information for curriculum development and student activities. In Hong Kong, the Education Bureau (2016) put forth the Promotion of STEM Education: Unleashing Potentials in Innovation, as a blueprint for curriculum reform tailored toward the STEM fields. And in Australia, the government has committed large sums of resources to boost student interest and involvement in STEM subjects (Department of Education and Training 2014).

Many of these initiatives have recognized that STEM education is a systematic process, and that future STEM talents cannot be trained overnight by one program. Researchers use the STEM pipeline framework to depict this progressive pathway of attracting students into STEM fields, obtaining relevant degrees, and eventually working in STEM careers (Sass 2015; Chen and Kelly 2013; New York Hall of Science 2012). The pipeline metaphor contributes to our understanding of STEM education; a successful STEM education requires a systemic and sustained effort throughout every stage of an education system. Although people in the field debate on the exact starting point of the STEM pipeline, there is no denying that early educational exposure correlates with the retention of STEM talents.

As the cornerstone for STEM education, mathematics achievement is closely linked to future STEM education (Wang 2013). Gender group difference in math achievement has been considered as one possible factor for the observed gender gap in the STEM pipeline (Sass 2015). Research shows that enrollment in calculus during high

school is associated with a higher level of performance in introductory college science courses, typically called gateway courses, which are often required for those expecting to choose STEM-related majors (Sadler and Tai 2007). Research evidence appears to suggest that students with the greatest likelihood of entering the STEM fields will typically come from the upper tail of the math performance distribution (Fan et al. 1997). It is also well-known that many STEM-related programs (e.g., physics, computer science) in competitive universities will only consider applicants with a very strong record in math achievement. From this perspective, gender mean difference in math achievement may only be marginally relevant, because students with an average performance in math typically would not, or would not be able to, choose STEM majors or careers.

Research Questions

In light of the alternative perspective on the gender gap in mathematics and concerns for STEM educators, in this study, we intended to examine the following two main issues:

1. What are the distributional differences between male and female students among the high achievers in math performance?
2. What socio-demographic and schooling/attitudinal variables are associated with the gender group differences among the high achievers in math performance? Does such a gender gap change after controlling for these background variables?

The current study used PISA 2003 and 2012 data to examine the gender gap among high achievers in math. After controlling for some potentially relevant socio-demographic, schooling, and attitudinal variables, the relevance of a gender gap among math high achievers for the STEM pipeline is discussed. For a better understanding of this issue within the international context, we examined several countries/regions across different continents.

Methods

We used data from the Program for International Student Assessment (PISA). PISA is one of the largest international assessment programs. It is organized by the Organization for Economic Co-operation and Development (OECD) and carried out on a three-year cycle. In 2012, PISA assessed the skills and knowledge of 15-year-old students who were enrolled in schools in 65 education systems (countries/regions) at the time. Three academic subject areas were assessed: reading, mathematics, and science. We used the math component of the PISA assessment of 2012 and the corresponding achievement scores.

In addition to the fact that the PISA 2012 data are the most recent data available for analysis[1], there is another important advantage to using this assessment for examining performance in mathematics. To reduce the burden of long testing sessions, PISA rotates its main assessment domain in each cycle. The 2012 cycle is the most recent assessment that specifically focused on mathematics, rather than reading or science. With mathematics as the main assessment domain, researchers have a wider selection of items to calibrate students' skills and knowledge of mathematics. Compared to the previous cycles in 2009 and 2006, the 2012 assessment for mathematics achievement has a higher level of precision, which is crucial for identifying the high-achieving student population in mathematics. To provide a historical comparative perspective, we also used PISA 2003 data, which was the last time mathematics was the measurement focus of PISA. The data from these two assessment cycles enabled a historical comparison of the mathematics assessment over a 10-year period.

In addition to the assessment of the three subject areas (reading, math, and science), information on student background and school environment were also available in the data. The statistical/analytical issues related to PISA data (e.g., purposeful oversampling) were accounted for in the current analyses by using appropriate weights as provided in the manuals for the PISA data (OECD 2009).

For understanding this issue within wider international contexts, ten education systems (countries, regions, economies) were selected for representing North America (USA, Canada), Europe (France, Germany, Italy), Asia (Hong Kong, Korea, Japan), South America (Brazil), and Oceania (Australia), and parallel analyses were conducted across these ten countries (regions). Descriptive analyses for the gender performance gap in math, proportional analysis for the gender gap among the high achievers in math, and logistic regression analysis for controlling for socio-demographic and schooling/attitudinal variables were conducted to elucidate the issues of our research interest.

[1] As of this writing, PISA 2015 data have just become available. Because PISA 2015 assessment did not have mathematics as the main assessment, but had science as the main assessment instead, for the reasons explained in the text, we decided not to analyze 2015 PISA data for our study.

Results

Descriptive Comparisons

As shown in Figure A1, there is a small gender mean difference in all ten countries/regions in 2003 and 2012. The effect size[2] for the gender group difference is typically very small across the ten different countries/regions. The largest effect sizes (e.g., South Korea, Italy, Brazil) are around 0.20 or below, and a few countries/regions have effect sizes around 0.10 and below (e.g., USA, France, Australia). As is widely recognized in educational research practice, effect size (*d*) of 0.20 is characterized as a small effect size (e.g., Cohen 1988) that represents a small group difference. For the US sample, the effect size for gender group difference is around 0.05, an almost negligible difference between 15-year-old male and female students. This finding (i.e., a very small gender difference in average math performance) is generally consistent with the majority of the recent existing literature that shows that male students have a very slight advantage in math performance over female students. According to OECD's own account, girls only outperformed boys in five out of 74 participants in the PISA 2012 cycle (OECD 2014a).

We replicated the analysis using the PISA 2003 data when mathematics was also a major domain of assessment. From a longitudinal perspective, the gender difference, although very small in some countries (e.g., USA) persists from 2003 to 2012. In Hong Kong and Japan, there is an increasing overall male advantage in mathematics. The same change applies to Australia, Brazil, and Germany, although to a lesser degree.

Gender Gap Among High Achievers in Math

The gender group mean difference (or the lack thereof) may only be marginally relevant for understanding the gender gap in STEM education, because average performers in math typically would not choose, or would not be able to choose, STEM careers late in life. High achievers in math in secondary schools, on the other hand, are more likely to enter STEM careers. For understanding the gender differential in the STEM pipeline, it is important to examine the composition of high achievers in math. Extending the earlier results from Guiso et al. (2008) and Stoet and Geary (2013), Fig. 1 displays the gender gap (in the form of male-to-female ratios) above progressively higher math performance levels (those scoring above the 50th, 75th, 90th, and 95th percentiles) by country/region.

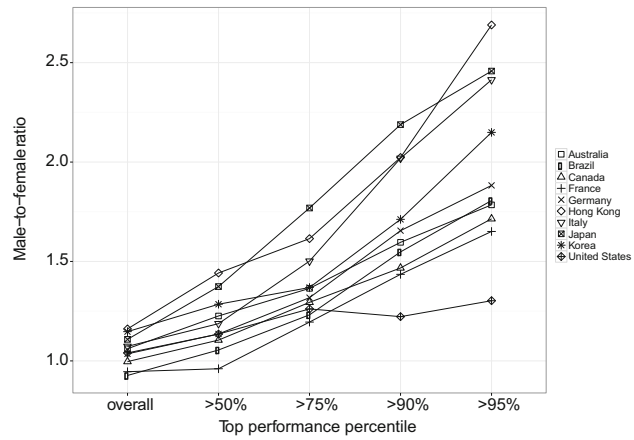

Fig. 1 Male-to-female ratio in overall sample and in top-performing groups

These estimated male-to-female ratios at the higher end of the performance distribution provide an informative picture about the gender gap among the high achievers in math. While the gender gap does seem trivial from the mean difference perspective (Figure A1), among the equally able high achievers in math, gender difference in the form of gender ratio could be substantial. Further, the more extreme the performance level is, the larger the male-to-female ratio. This trend is consistent for nine out of the ten country/regions examined, with the only exception being the United States. In the United States, though the male advantage at the higher end of the ability distribution holds true, the trend does not seem to change when we focus on the highest-performing students. To verify the robustness of these findings as shown in Figure A1 based on the 2012 PISA data, we replicated the same analysis with the PISA 2003 data, the previous PISA cycle when mathematics was the major domain of assessment, and the results (not presented here) were very similar.

Figure 1 shows that the gender gap at the higher end of the math performance distribution clearly exists.[3] To consider Canada as an example, the male-to-female ratio for the total sample is 0.996 (i.e., 0.4% fewer male-to-female students in the overall sample). But among those scoring above the 50th, 75th, 90th, and 95th percentiles of math assessment, the male-to-female ratio was progressively larger: 1.10, 1.29, 1.47, and 1.71, respectively. In other words, among the top 10 percent of performers on the PISA

[2] The effect size is the standardized mean difference in math performance, as represented by Cohen's d $\left(d = \frac{Mean_{male} - Mean_{female}}{S_{pooled}}\right)$. Five effect sizes were calculated based on five plausible values, and then we took the average of these five.

[3] For simplicity, we did not present inferential statistics of the ratio estimates in the figure. We did compute the 95% confidence interval for all the ratio estimates. The method and results are enclosed in the Appendix. Nine out of ten countries/regions have statistically significant male-to-female ratios higher than 1, with the United States being the exception.

2012 math assessment, there were close to 150 male students for every 100 female students.

The most extreme case in our sample of countries/regions is Hong Kong. In Hong Kong, the male-to-female ratios among those in the top percentile groups (50th, 75th, 90th, and 95th percentiles) of the math assessment are 1.44, 1.61, 2.20, and 2.69, respectively. These startling numbers suggest, for example, that among the highest-performing 10 percent of students, for every 100 girls, there are 220 boys. Similar findings apply to Italy and Japan to a slightly lesser degree. Such a gender gap among high achievers in math could have serious implications for the STEM pipeline.

The findings of substantially more boys performing at higher levels of math achievement than girls are in agreement with OECD's own calculations based on a different metric. Among OECD countries, on average, 14.7 percent of boys reached proficiency level 5 or above,[4] whereas 10.6 percent of girls performed at the same level (OECD 2014a).

Interestingly, the United States is the only country where the 95% confidence interval of gender ratio estimates include 1, indicating that, although the observed gender ratio shows a male advantage among the high achievers (1.22 for the top 10 percent and 1.30 for the top 5 percent), the estimation uncertainty is large enough such that the ratio (i.e., 1.30) is not statistically different from 1 (equal proportions of male and female students). This finding could be considered from two different perspectives. First, there could be too much "noise" in the estimation due to the relatively smaller number of students in the high end of the performance distribution. We examined the 2003 US PISA data on this issue, and the results showed that the male-to-female ratios among the top 10 and top 5 percent of math performers were 1.54 (CI 1.11, 1.97) and 1.94 (CI 1.15, 2.73), respectively, which aligned with the findings from other countries.

Second, while the information about male-to-female ratios among the top math performers in the US was encouraging, indicating that the gender group difference at the higher end of the math performance distribution is the smallest among the ten countries/regions examined, the overall performance of US students in the PISA 2012 math assessment was discouraging. Figure 2 presents the country/region-specific scores corresponding to the four upper percentile points (50th, 75th, 90th, and 95th) for the ten countries/regions included in this study. While Brazil had the lowest performance scores among the ten countries/regions, the US is the second from the bottom overall (approximate order of performance: Hong Kong, South Korea, Japan, Germany, Canada, Australia, France, Italy, USA, Brazil), and has the lowest performance among the six developed western nations (approximate order of performance: Germany, Canada, Australia, France, Italy, USA). The performance difference among the ten countries/regions is striking in some cases. For example, the difference between the lowest-performing country/region (Brazil) and the highest-performing country/region (Hong Kong or South Korea) is about 185 points, representing approximately 1.8 standard deviations on the PISA math assessment scale. This would be considered a huge difference by almost any standard.

The difference between the United States and the top-performing country/region (Hong Kong or South Korea) is also very large at about 80–90 points, representing approximately 0.8–0.9 standard deviations on the PISA math assessment scale. Viewed from another perspective, the score for the 90th percentile in the US sample was 600, which was lower than the score for the 75th percentile of three countries/regions (Hong Kong: 629; South Korea: 624; Japan: 603). In other words, the US "elite" top 10% math performers would be considered much less elite when compared with the better performing countries/regions, as the scores of the US top 10% were obviously lower than the top 25% of the high performer groups in these three countries/regions. Accordingly, the data here reveal that, although the gender gap among the top math performers in the US is less pronounced, unfortunately, this smaller gender gap is based on a much lower overall math performance of the US students.

PISA data also offer another angle for understanding the gender gap issue. In addition to the overall achievement score in mathematics, the 2012 assessment provides detailed subscale scores. Specifically, there are two broad areas: mathematical content and mathematical process. The

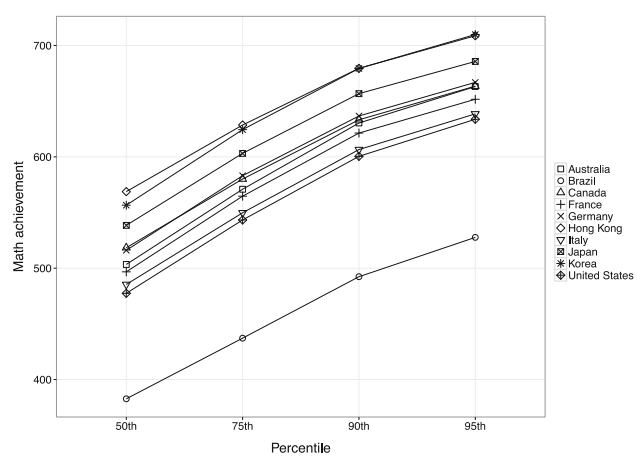

Fig. 2 Scores corresponding to four upper percentile points across ten countries/regions

[4] Proficiency level is OECD's way of categorizing student performance. Proficiency level ranges from the lowest level 1 to highest level 6. Only 3.3 percent of OECD students achieved level 6 in mathematics in 2012. For details, see OECD (2014a, p. 61).

content subscales include *change and relationship*, *space and shape*, *quantity*, and *uncertainty and data*. The process subscales include *formulating*, *employing*, and *interpreting*. Each of these subscales measures distinctive domains of mathematical knowledge, and they were validated by a group of international expert panels (OECD 2003, 2013). Just as the overall achievement scale, these subscales have the same international mean and standard deviation. For each economy, we computed the male-to-female ratios above the selected percentiles at the upper end of performance distributions. For the sake of simplicity, Fig. 3 presents the male-to-female ratios among the top 10% and top 5% of high performers for each subscale.

Within each country/region, the pattern is incredibly consistent: there are more boys than girls at the right extreme upper end of the performance distribution (i.e., male-to-female ratio >1); the more extreme the performance is, the bigger the gender gap. It also shows that the gender gap is persistent across a wide range of subdomains of mathematical skills and knowledge among top performers. The gap is not driven by one particular area; instead, the gap exists in all areas. Again, the gender gap for the US sample is the smallest among the countries/regions examined here.

Gender "Effect" After Controlling for Background Variables

For many educators and policy makers, there is a pressing question about gender gaps such as the one shown in Fig. 1: could such a "gender" gap be attributed to socio-demographic and schooling/attitudinal variables, rather

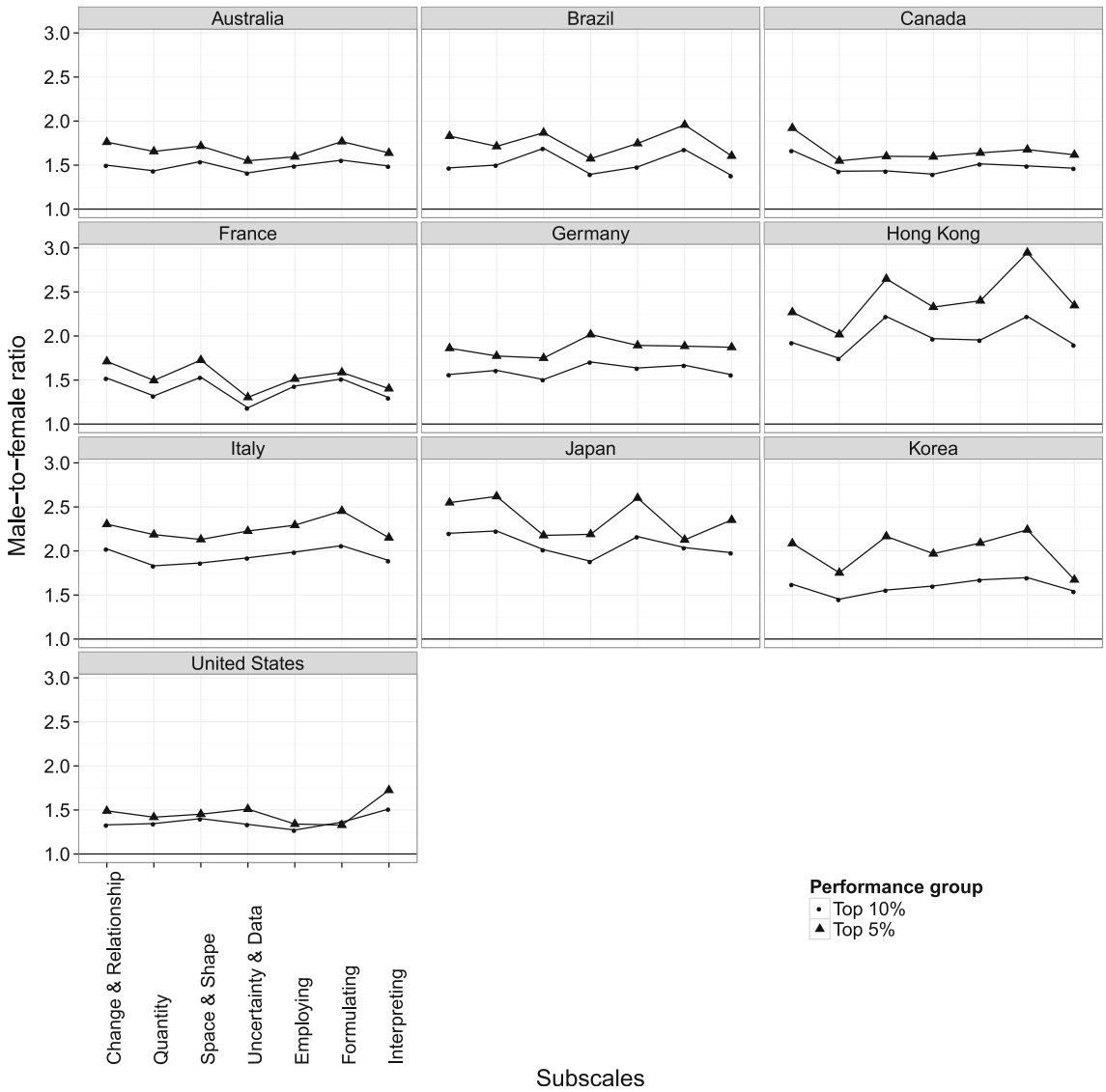

Fig. 3 Male-to-female ratio among students performing above 90th and 95th percentiles on seven subscales of mathematics, PISA 2012

than to gender itself? In other words, could the "gender" gap actually represent the difference contributed by some other variable(s) that happen to coincide with the gender status? To examine this question, we used a group of socio-demographic and schooling/attitudinal variables (*out-of-school lesson*, *math self-efficacy*, *math anxiety*, *attitude toward school*, and *SES*: *social and economic status*), together with "gender" to model the probability of being in the top-performing group (either scoring above the 90th, or above the 95th percentile of math score distribution). The main objective was to examine if "gender" would still be related to the probability of being in the top-performing group after controlling for these socio-demographic and schooling/attitudinal variables.

Including these background variables in the analysis model resulted in an analysis complication. For the purpose of reducing the burden of the student respondents, the PISA 2012 student questionnaire adopted a rotating design, which divided the student background questions into multiple booklets, and rotated these booklets among the sampled students (OECD 2014b). As a result, students were not able to respond to all the available questions. Consequently, because the students answered different subsets of questions, the rotating design made it impossible to control for *all* the individual characteristics simultaneously in one analysis model. More specifically, for the five background variables that we considered (see description above), *SES* was available for all sample members, but only one of the four (*out-of-school lesson*, *math self-efficacy*, *math anxiety*, *attitude toward school*) was available for a particular portion of a country/region's sample. For this reason, to predict the membership in each top-performing group (either >90th percentile, or >95th percentile), we ran four logistic regression models, with each model controlling for *SES* together with one of the four other background variables mentioned above. This approach resulted in producing four odds ratios (ORs) for the variable "Gender" shown in Table 1 for predicting the membership probability in a top-performing group (>90th percentile, or >95th percentile).

Controlled Background Variables

The controlled background variables were used so that we could have a better estimate for gender group difference related to the upper tail (>90th and 95th percentiles) of performance distribution. For this reason, for the sake of presentation simplicity, the odds ratios (ORs) for the controlled background variables were not presented in Table 1.[5] However, the results concerning these background variables were generally very consistent across the countries/regions. Here we summarized the general findings related to these controlled background variables. Consistently, *SES* was a strong and statistically significant predictor for membership in both top-performing groups (i.e., >90th and 95th percentiles) across the countries/regions, with ORs typically being in the range of 1.5–3.5. *Out-of-school lessons* (that is, whether and how often a student took supplemental lessons outside of school) was typically a negative predictor (i.e., OR < 1) for membership in the top-performing math group (i.e., the more out-of-school lessons a student took, the less likely that he/she would be in the math top-performing group), except for Japan and South Korea where the opposite was observed (OR > 1). *Math self-efficacy* and *math anxiety* are statistically significant and strong predictors for membership in the top-performing group of math, with *math self-efficacy* being a consistently positive predictor (OR $\approx$ 2–3; those reporting a higher level of *math self-efficacy* were much more likely to be in the two top-performing math groups), while *math anxiety* was a consistently negative predictor (OR $\approx$ 0.5; those reporting a higher level of *math anxiety* were less likely to be in the two top-performing math groups). *Attitude toward school*, in general, turned out to be a statistically non-significant predictor for membership in the top-performing math groups.

Gender "Effect"

Table 1 presents the odds ratio (OR) of gender (male = 1, female = 0) for predicting membership in the two top-performing math groups *after* controlling for the background variables discussed above, with OR > 1 indicating that male students were more likely to be in the two top-performing math groups than female students.[6] These results would help us to understand if the observed gender group "gap" in the top-performing math groups as shown in Fig. 1 would still hold after controlling for some relevant background variables.

Table 1 shows that, after controlling for *SES* and other attitudinal/psychological variables (*out-of-school lesson*, *math self-efficacy*, *math anxiety*, or *attitude toward school*), the gender "effect" remained in most situations. In other words, after controlling for the background variables, in general, male students were still more, sometimes substantially more, likely to be in the top-performing math groups (>90th percentile; >95th percentile). This "gender effect" was more obvious in some countries/regions than in others. For example, for the samples from Hong Kong,

[5] Detailed information about the *OR*s of the background variables is available upon request from the corresponding author.

[6] To account for design effect of complex sampling, we applied statistical weights at both the student and school level. The weights were scaled using a method proposed by Pfeffermann et al. (1998).

Table 1 Odds ratios[a,b] (OR) of "gender" for predicting membership in two top-scoring groups across ten countries/regions after controlling for background variables

	Above 90th percentile				Above 95th percentile			
	Models[c]				Models			
	A	B	C	D	A	B	C	D
Australia	1.58	0.91^	1.18^	2.42	1.77	0.97^	1.15^	1.51
Brazil	1.65	1.30	1.43	2.73	1.75	1.43	1.49	1.88
Canada	1.49	1.10^	1.21^	1.94	1.68	1.23^	1.44	1.96
France	1.46	1.03^	1.24^	3.5	1.66	0.98^	1.26^	2.11
Germany	1.57	0.97^	1.27^	2.24	1.83	1.05^	1.38^	1.75
Hong Kong	1.84	1.31^	1.61	1.67	2.35	1.85	2.09	2.58
Italy	2.04	1.50	1.81	1.88	2.43	1.70	2.02	2.37
Japan	1.92	1.34	1.97	2.64	2.19	1.44	2.27	2.49
South Korea	2.01	1.63	1.62	2.86	2.21	2.01	1.87	2.2
USA	1.19^	0.92^	1.19^	2.62	1.21^	0.93^	1.34^	1.50^

[a] Odds ratio (OR[a,b]) for gender (M = 1, F = 0). $OR = 1$ indicating that gender has no predictive power for membership in the top-performing group. $OR < 1$ indicates that males are less likely to be in the top-performing group (above 90th percentile, or above 95th percentile); $OR > 1$ indicates that males are more likely to be in the top-performing group. [Odds ratios (OR[a,b]) for Gender: M = 1, F = 0]

[b] All *OR*s are statistically different from 1 at $p = .05$ or below, except those with "^"

[c] For assessing the "gender effect" for predicting membership in each top-performing group (>90th percentile, or >95th percentile), four logistic regression models were conducted in order to control for the socio-demographic and schooling/attitudinal variables. Model A controls for "Out of school lesson" and SES; Model B controls for "Math self-efficacy" and SES; Model C controls for "Math anxiety" and SES; Model D controls for "Attitude toward school" and SES. This is necessitated by the rotating sampling design of PISA student survey described in the text (under the subheading *'Gender "Effect" after Controlling for Background Variables'*) whereby each respondent only responded to one of the four background/schooling variables. As a result, four ORs were presented for variable "gender" for predicting one top-performing group membership

Italy, Japan, and South Korea, male students were about twice as likely (OR ≈ 2 or above) than their female counterparts to be in the elite top 95th percentile group in math performance, even after controlling for students' *SES* and one other attitudinal/psychological variable. Similar, but less strong, trends were observed for other countries/regions. Consistent with Fig. 1 where the USA sample showed the smallest male advantage (i.e., lowest male-to-female ratios in favor of male students) in the top-performing math groups, as shown in Table 1, after controlling for the background variables, the USA sample also showed the lowest ORs (in favor of male students) for "gender" to predict membership in the top-performing math groups. In addition, most of the ORs for the United States sample were not statistically significant.

Discussion

The STEM fields have historically been male dominated (Ceci and Williams 2011). Many countries are currently acting to reduce the gender inequality from the STEM pipeline, but a substantial gender gap remains. Take the United States as an example, where the relevant statistics are more readily available. For all jobs in the nation, men's and women's shares of the workforce are 52 vs. 48% respectively. But for STEM jobs, the men's share is 76%, and the women's share is only 24% (Department of Commerce 2011). Such a huge gender differential in STEM careers has occurred despite favorable wages where the earnings for women with STEM jobs were 30% higher than for comparable women in other industries. Furthermore, the wage premium of STEM jobs for women was considerably higher than the STEM premium for men. Despite these tangible benefits for women in STEM careers, the gender differential in the STEM workforce stubbornly persists, and has remained unchanged for over a decade (the women's share of the US STEM workforce was 24% in both 2000 and 2010) (Department of Commerce 2011).

Undoubtedly, there must be a multitude of factors contributing to this situation (e.g., gender difference in math learning, lack of female role models for girls, societal/cultural gender stereotyping, social/psychological affective factors) (Geary 2011). In this study, we probed into one possible contributor of the gender imbalance in STEM: gender disparity among mathematics high achievers in the form of gender ratios. Our analyses show that gender disparity among high achievers is a global issue, and that it becomes more acute when students move up the

performance distribution in mathematics. Additionally, the pattern of disparity remains similar in both 2003 and 2012, and the gender disparity also penetrates the subdomains of mathematics, resulting in the lopsided distribution pattern of the two genders in different areas. Gender disparity is the largest in Hong Kong and Japan. Policy makers in these education systems should be aware of the gender imbalance regarding the STEM pipeline at the secondary levels.

Our analysis also suggests that math self-efficacy was closely associated with a high level of performance in math. Although one's math self-efficacy could have been influenced by one's performance in math, the reciprocal effect of self-efficacy on performance is also well recognized. A large body of research literature on self-efficacy has shown that one's self-efficacy can have a powerful influence on an individual's behavior and success (e.g., Bandura 1997; Benight and Bandura 2004) in general. Similar findings abound in academic learning research (e.g., Bandura et al. 1996; Pajares and Margaret 1994). The research in STEM areas also confirmed the positive, and often powerful, effect of self-efficacy. For example, Betz and Hackett (1986) showed that young women's career-related behaviors and career choices in STEM areas were strongly influenced by the gender differences in self-efficacy expectancies. Furthermore, for general math interest and other math-related choices (e.g., math courses, and math majors), math self-efficacy proved to be a more powerful predictor than past math achievement itself (Pajares 1996). Research in technology learning also suggested that the effect of technical self-efficacy could be a stronger predictor for better technology learning outcomes than the effect of gender (Brauner et al. 2010), despite the very strong stereotypical perception that males had an overwhelming advantage in technology learning and careers.

The empirical research literature clearly indicates that high math achievement is quite likely a product of both cognitive ability and a student's self-efficacy related to math (Geary 2011; Rohde and Thompson 2007; Parker et al. 2014). Self-efficacy, however, is a trait that is prone to intervention and modification. The implication for educators is clear: "… school practitioners should look to students' self-beliefs about their academic capabilities, for they are important components of motivation, self-regulation, and academic achievement" (Pajares 2002). More specifically, math education should aim to cultivate math self-efficacy and confidence in girls, especially at younger ages. As discussed by Bandura (1997), self-efficacy could be influenced by multiple factors, such as performance accomplishment, vicarious experience, verbal persuasion, and physiological states. The challenge for educators is to find ways to create relevant and authentic situations in which young girls may positively benefit from these factors.

Lastly, we note that although the hypothesis of "gender similarities in math achievement" (Hyde 2005; Hyde et al. 2008; Hyde and Mertz 2009) has received much attention within the education and science communities, it is primarily based on empirical evidence of gender groups' mean difference (or the lack thereof) in math achievement. Also, by only focusing on the United States, a possible outlier revealed by our analysis, the whole picture of learning outcomes by gender groups on the global stage was not accurately shown. The phenomenon of a small or even non-existent gender mean difference represents the tremendous social progress that has been made for gender education equity. However, the hypothesis of "gender similarities in math achievement" as it is currently conceptualized and operationalized easily leads to the view of a gender-balanced STEM pipeline. We argue that it is insufficient for understanding the gender differential in the STEM pipeline. Particularly when we look at the international picture, there is an obvious gender distributional gap among the top performers in math, and this could be one viable factor contributing to the persistent gender differential in the STEM pipeline.

The persistent gender gap among the top performers of math achievement (Fan et al. 1997; Guiso et al. 2008; Hyde et al. 2008; Stoet and Geary 2013) presents a clear challenge to society and the math education profession. It is not enough to just reduce or eliminate the gender mean difference. Yet, our understanding of the factors that potentially contribute to such an imbalance has just begun to emerge (Stoet and Geary 2015). We call on future research to further understand the causes of the gendered distribution of mathematical performances.

To empower and to enable more female students to consider and enter STEM careers, we must continue to address the huge challenge to close the gender gap among the high achievers in math. Only then may we be able to claim success in having achieved true gender similarities and equality in math achievement.

Limitations

This study uses cross-sectional data from two waves of PISA. Though we were able to compare 10 developed and developing nations/economies from the globe, we have not extended the comparisons to every country that has ever participated in PISA. To generalize the results of the current study requires replication in other countries. PISA's cross-sectional nature also determines that the multivariate results we found are associational and should be interpreted as causal.

Conclusion

In this study, we probe into gender inequality in mathematics from an alternative angle: gender distributional imbalance among high achievers. We contribute to the current literature by moving away from mean score comparison to distributional differences. Using two waves of PISA data from ten countries, our analysis shows persistently fewer female achievers compared to males at age of 15. Gender disparity became more tilted when students move up the performance distribution. In regression analysis, even after controlling for a series of factors related to mathematics achievement, the gender disparity persists. Our preliminary analysis shows higher self-efficacy is correlated with lower gender imbalance in several education systems. Future studies are suggested to examine the causal nature of what matters to mitigate gender imbalance for the preparation of STEM talents.

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